



Department of Electronic and Computer Engineering

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Final Project Report (Summative)

Construction Site Robotic Arm

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ABSTRACT

The Construction Site Robotic Arm is an Internet of Things (IoT)-based automated system designed to assist in picking and placing construction materials such as bricks. The project integrates an ESP32 microcontroller, servo motors, and an ultrasonic sensor to detect, pick, and place objects within a defined range. The system operates in two modes: manual and automatic. In manual mode, users control the arm remotely using the Blynk mobile application through Wi-Fi communication, allowing movement of individual joints such as the base, shoulder, elbow, and gripper. In automatic mode, the ultrasonic sensor detects an object within 10 cm and triggers a predefined sequence of servo movements for object pickup and placement. The robotic arm includes indicator LEDs that display the operational states of the system, such as automatic mode, manual mode, and object detection. Power is supplied through a regulated 5V external source shared among all servos, ensuring stable operation. The system demonstrates a practical implementation of IoT in automation and robotics, enhancing productivity, precision, and safety in construction sites. This project serves as a foundation for future developments in smart construction, where robotics and IoT can work together to reduce human labor and improve efficiency.

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List of Abbreviations

Abbreviations	Meaning
DC	Direct Current.
DOF	Degrees of Freedom.
ESP32	A low-cost microcontroller with Wi-Fi and Bluetooth capabilities.
FARA	Framing Assembly Robotic Arm.
GPIO	General-Purpose Input/Output (pins on a microcontroller).
GUI	Graphical User Interface.
HC-SR04	Ultrasonic distance sensor.
IoT	Internet of Things.
LDR	Light Dependent Resistor.
LED	Light Emitting Diode.
MG996R	A high-torque servo motor model.
PWM	Pulse Width Modulation (used for servo motor control).
RGB-D	Red-Green-Blue-Depth (camera technology, e.g., Microsoft Kinect).
ROM	Read-Only Memory.
SRAM	Static Random-Access Memory.
TTL	Transistor-Transistor Logic (a digital signal standard).
Wi-Fi	Wireless Communication Protocol (wireless networking technology).

CHAPTER 1: INTRODUCTION

Automation, remote operation, and safety are all becoming increasingly important in current construction operations. Traditional manual work for carrying and installing big materials like bricks can be time-consuming, physically taxing, and dangerous in certain situations. Robotic systems are increasingly widely employed in industries such as automobile production, logistics, and construction to execute repetitive or dangerous jobs with accuracy, consistency, and efficiency. This project introduces a Construction Site Robotic Arm, which is intended to simulate similar industrial automation on a smaller, instructional size. It employs an ESP32 microcontroller with Wi-Fi connectivity, allowing it to be operated wirelessly using the Blynk smartphone application. The robotic arm is equipped with an ultrasonic sensor to detect bricks and servo motors for picking and placement [1].

This project employs a variety of modern technologies that reflect real-world industrial automation systems. The system is built on the ESP32 microcontroller, which provides both processing power and wireless connectivity. It connects to the internet via Wi-Fi and communicates with the Blynk IoT platform, enabling for remote control and monitoring of the robotic arm using a smartphone app. The Blynk interface functions as a digital control panel, allowing users to choose between manual and automatic modes and control arm movements wirelessly. An ultrasonic sensor (HC-SR04) is used to detect the presence of bricks in front of the arm, allowing for autonomous operation when in automated mode. Servo motors are utilized to control the movement of the arm, which includes rotating, lifting, and gripping, enabling precise mechanical control similar to that seen in industrial robotic arms. These technologies show how IoT, automation, and robotics may be used to build a smart system for construction and material handling applications [2].

This project models an automated material-handling system used in construction. In automatic mode, the arm recognizes and lifts bricks without the interaction of the user. In manual mode, the Blynk app allows the user to remotely control the movements of the robotic arm. The system highlights how IoT and robotics may be combined to increase efficiency, flexibility, and worker safety in industrial and construction environments [3].

1.1 PROJECT OBJECTIVES

1. Develop a robotic arm system for picking and placing bricks in a construction site environment.
2. Use an ESP32 microcontroller to control the robotic arm and establish wireless connectivity using the Blynk IoT platform.
3. Use an ultrasonic sensor to detect surrounding things, such as bricks, and automate the picking process.
4. Enable dual-mode operation:
 - Automatic Mode: The system recognizes and transfers bricks automatically.
 - Manual Mode: Allows users to remotely control the arm via the Blynk app.
5. Use servo motors to accurately handle objects with the robotic arm's rotation, lifting, and gripping capabilities.
6. Develop a user-friendly interface on the Blynk app for controlling the robotic arm, viewing system status, and switching between operation modes.
7. Demonstrate how IoT and automation may improve safety, efficiency, and precision in real-world industrial and construction applications [4].

1.2 Applications, benefits and limitations of Construction Site Robotic Arm

1.2.1 Applications

- **Brick Lifting and Placement** – Automates the repetitive task of picking up and placing bricks on construction sites.
- **Remote-Controlled Material Handling** – Operators can control the arm from a safe distance with the use of a smartphone.
- **Industrial Training and Education** – Useful in technical training institutions to demonstrate robotics, IoT, and automation.
- **Smart Construction Sites (Future Use)** – Can be integrated into IoT-based smart sites for automated material movement.
- **Prototype for Larger Automated Systems** – Serves as a base for developing fully autonomous construction robots [5].

1.2.2 Benefits

- Reduces manual labor by eliminating repetitive physical duties for workers.
- Improves safety by preventing workers from performing unsafe lifting jobs.
- Remote control via Blynk eliminates the requirement for close physical contact.
- Cost-Effective Prototype: Designed with economical components for education and experimentation.
- Customizable and Scalable: Upgrade with more sensors, AI, or machine learning for further automation.

- Enhances technological skills through microcontroller programming, electronics, robotics, and IoT. Ideal for learning [6].

1.2.3 Limitations for the project

1. Limited to lift any mass above 0.1–0.15 kg (100–150 g) due to the torque limitations of the SG90 servos, it can only lift lightweight goods.
- Torque Calculation Formula was used to identify the safe load

$$\tau = F \times d \quad [1-1]$$

Where:

τ = torque of the servo (kg·cm)

F = force applied by the load (kg)

d = distance from the servo axis to the load (cm)

2. Limited Sensing Capabilities - Requires a single ultrasonic sensor and cannot detect different object types or environments.
3. No Obstacle Avoidance
4. Not weatherproof
5. Manual Calibration Required -Precise placement requires human calibration, as there is no feedback system for position correction.
6. Limited to the Wi-Fi signal region.

CHAPTER 2: LITERATURE REVIEW

The development of robotic arms for automation has been extensively studied in a variety of fields, including industrial manufacturing, smart systems, and construction applications. This chapter examines three significant studies that provide insights into the design, actuation, control methods, and application of robotic arms, with a particular emphasis on IoT integration, servo actuation, and construction-site deployment. The evaluation identifies existing successes and research gaps that the current initiative addresses.

2.1 IoT and Microcontroller-Based Robotic Systems

H. Ahmad et al. [6] demonstrated an IoT-enabled automation system based on the ESP32 microcontroller, ultrasonic sensors, and servo motors coupled with the Blynk IoT platform. Their results revealed the effectiveness of IoT in providing real-time monitoring and precise actuation, hence increasing total system accuracy and efficiency. The study found that servo motors are dependable actuators for robotic applications, and ultrasonic sensors provide accurate object detection in automated systems.

This work directly supports the design choices made in the current project, which also uses an ESP32, an ultrasonic sensor, and servo motors connected via the Blynk platform for remote operation. However, while the reviewed study concentrated on generic IoT automation, it did not address the unique issues of construction-site material management. The current project builds on this expertise by integrating these technologies into a robotic arm designed to detect, choose, and position construction bricks, bridging the gap between IoT-based control and construction automation.

2.2 Design and Construction of Robotic Arms

K. S. Sahu et al. [7] studied the building of a robotic arm with multiple degrees of freedom (DOF), focussing on servo-based actuation and mechanical design. The researchers showed that servos could reliably regulate base rotation, arm joints, and grippers, allowing the arm to perform simple pick-and-place tasks. Their findings also highlighted the need of strong mechanical design, including material selection and torque calculation, in ensuring structural stability and suitable load-bearing capacity.

While this study provides useful insights into the structural and actuation features of robotic arms, it does not include an IoT-enabled control framework. The current project improves on this foundation by not only using servo motors for arm movement but also incorporates IoT-based wireless control via ESP32 and Blynk, as well as ultrasonic sensing for object recognition. This approach modifies traditional robotic arm designs for use in modern construction-site automation applications.

2.3 Robotic Arms for Construction Applications

D. Lopez et al. [8] created the Framing Assembly Robotic Arm (FARA) system to address safety and efficiency issues on construction sites. The device included servo motors, infrared obstacle detection sensors, and a Microsoft Kinect RGB-D camera to improve situational awareness and prevent accidents. FARA successfully completed assembly tasks with a 3D-printed structural model, demonstrating the possibility of robotic systems for construction automation.

Although FARA demonstrated significant potential in construction, its reliance on manual controls limited its automation scope. The current project builds on these findings by implementing an IoT-enabled approach using ESP32, ultrasonic sensors, and Blynk cloud integration. This enhancement provides wireless monitoring, semi-automatic brick handling, and mobile-based control, thereby making robotic arms more adaptable to real-world construction scenarios.

CHAPTER 3: DETAILED PROJECT SPECIFICATIONS

3.1 System Operation

The Construction Site Robotic Arm detects, picks, and places construction bricks using a remote control and the Blynk app. The procedure begins with the ESP32 microcontroller connecting to the Blynk cloud platform over Wi-Fi, allowing the smartphone of the user to communicate wirelessly with the robotic arm. When turned on, the ultrasonic sensor at the base of the arm continuously scans for items within its detecting range. If an object, such as a brick, is detected within the 10cm distance, the system enters the "object found" state. Depending on the mode selected in the Blynk app, the user can control the arm manually or semi-automatically.

Three indicating LEDs (RED LED, YELLOW LED and GREEN LED) were used. YELLOW LED was used as Automatic Mode LED where was set via Blynk mode switch and was turned on when the arm is in Automatic Mode. RED LED was used as Manual Mode LED, it was controlled from the blynk buttons, and it was turned on when the arm is in Manual Mode. GREEN LED was used as Brick Detected LED and it was turned on when the ultrasonic sensor detected a brick within the pickup range of 10cm[9].

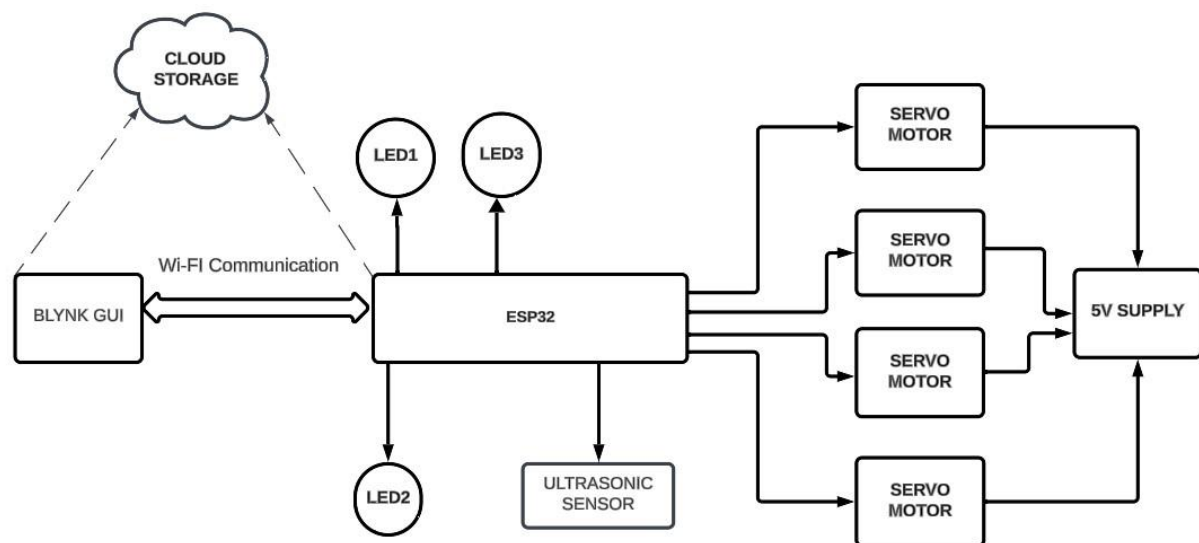


Figure 3. 1: System Block Diagram

In manual mode, the Blynk interface displayed on-screen control buttons or sliders that allow the user to independently operate each servo motor. This involves rotating the base, adjusting the arm joints, and using the gripper to pick up or release the brick. The operator may precisely position the arm to pick up the product and place it in the desired area, such as on a truck bed.

In automatic or assisted mode, once the ultrasonic sensor detects the brick, the ESP32 triggers a predefined sequence of servo movements:

- The base servo rotates towards the detected brick.
- The main arm servos adjust angles to align the gripper over the brick.
- The gripper servo closes to firmly grasp the brick.
- The arm lifts and rotates towards the placement area.
- The gripper releases the brick.

Throughout the process, the motor driver ensured that the servo motors receive the necessary voltage and current, while the ESP32 processed sensor data and control signals from the Blynk app. Power is supplied from a regulated supply that can drive several servos without voltage dips.

This operational flow made the robotic arm suitable for applications in small-scale construction sites, educational robotics, and industrial material handling, giving a practical example of IoT-enabled automation.

3.2 CIRCUIT DESIGN

The circuit in Figure 3.2 was created using Circuit Designer to control and automate a robotic arm for building purposes. The system includes an ESP32 microcontroller, an ultrasonic sensor (HC-SR04), four servo motors, and indicator LEDs. The ESP32 is the central processing unit, which receives data from the ultrasonic sensor, processes distance information, and generates control signals for the servo motors based on the operational mode selected. The ultrasonic sensor detects the existence of an object (brick) within a predetermined range, prompting the robotic arm to perform the pick-and-place operation.

Each servo motor has a distinct function: the base servo spins the arm horizontally, the shoulder and elbow servos lift and extend the arm, and the gripper servo opens and shuts to select and release the brick. The GPIO pins of the ESP32 receive PWM (Pulse Width Modulation) signals, which drive the servo motors. Indicator LEDs display system status: Automatic Mode, Manual Mode, and Brick Detected. A controlled external 5V DC power supply is provided to power the servo motors, which require more

current than the ESP32 can give. The grounds of the external power supply and the ESP32 are connected to establish a single reference point and steady operation.

The trigger and echo pins of the ultrasonic sensor are linked to GPIO 5 and 18, respectively. The servo motors are connected to GPIOs 13, 12, 14, and 27. LEDs are linked with 220 Ω current-limiting resistors to prevent excessive current consumption. This arrangement enables the robotic arm to make precise and synchronized movements in both manual and automatic modes.

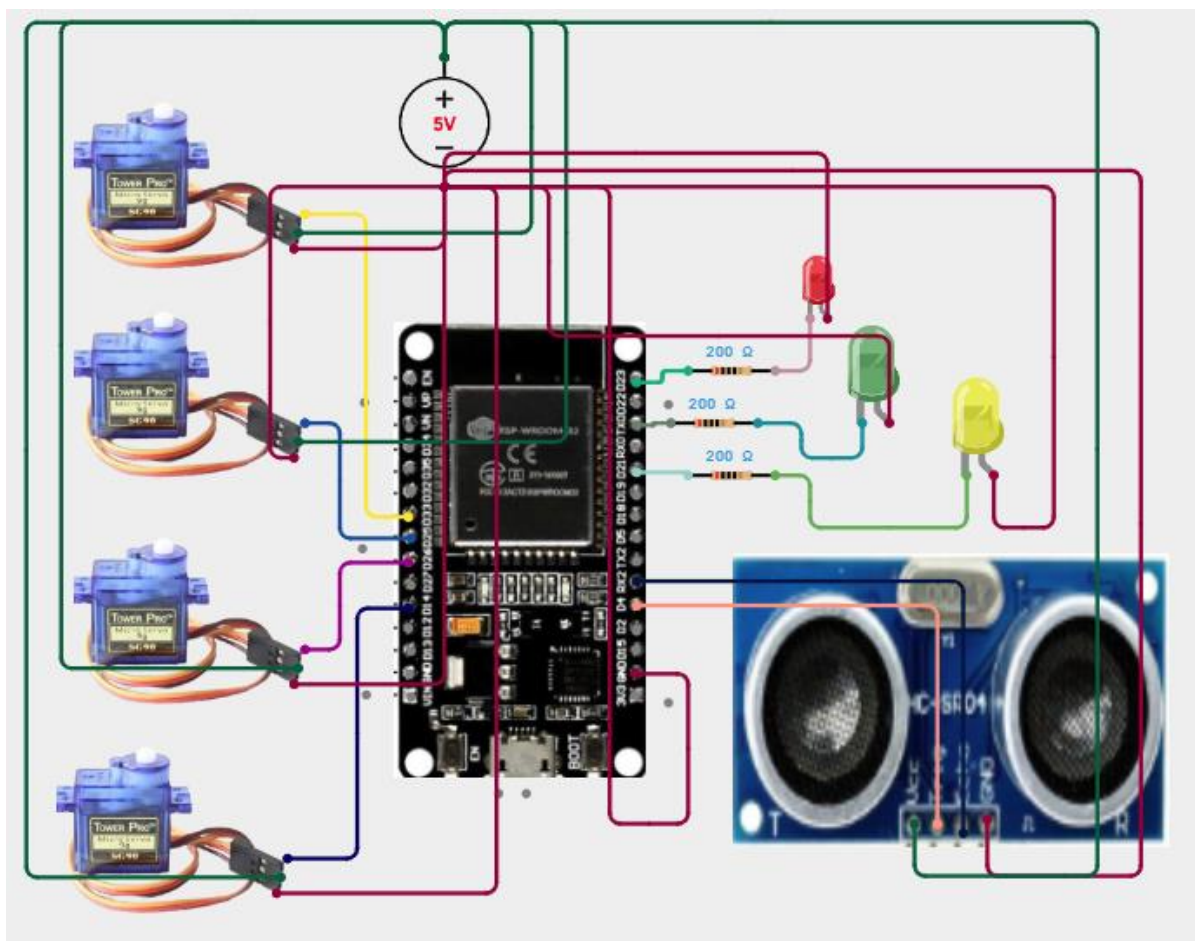


Figure 3. 2: System Circuit Diagram

3.3 GUI Design

The Graphical User Interface (GUI) was developed utilizing the Blynk IoT platform. The Blynk app works as a remote interface, allowing users to observe and control the robotic arm in real time. The dashboard features virtual buttons and sliders that connect to the ESP32 via the Blynk cloud server, allowing for bidirectional Wi-Fi communication. The interface is intended to give seamless manual control over the arm's movement and the ability to switch between operation modes.

In Manual Mode, the Blynk interface has separate control buttons for each servo motor, allowing the user to rotate the base, move the shoulder and elbow joints, and open and close the gripper. When the ultrasonic sensor detects an object within 10 cm, the system goes into Automatic Mode. To pick up and place the object, the ESP32 uses a pre-programmed series of servo motions. Indicator LEDs on the interface show the active mode and object detection status, giving the user real-time feedback.

This design enables the robotic arm to be wirelessly controlled and monitored, increasing flexibility, use, and safety during construction material handling tasks. The GUI design improves system interactivity by granting the user complete sight and control over the robotic movements of the arms from any connected device.

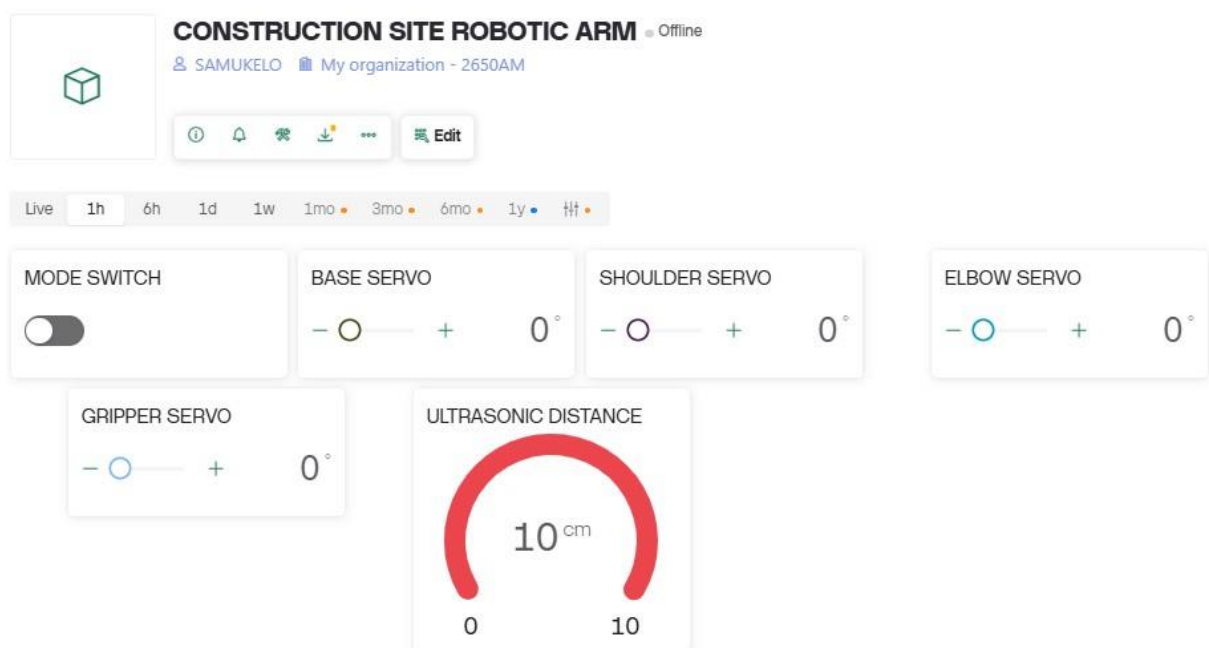


Figure 3. 3:GUI Design

3.3 FLOW CHART

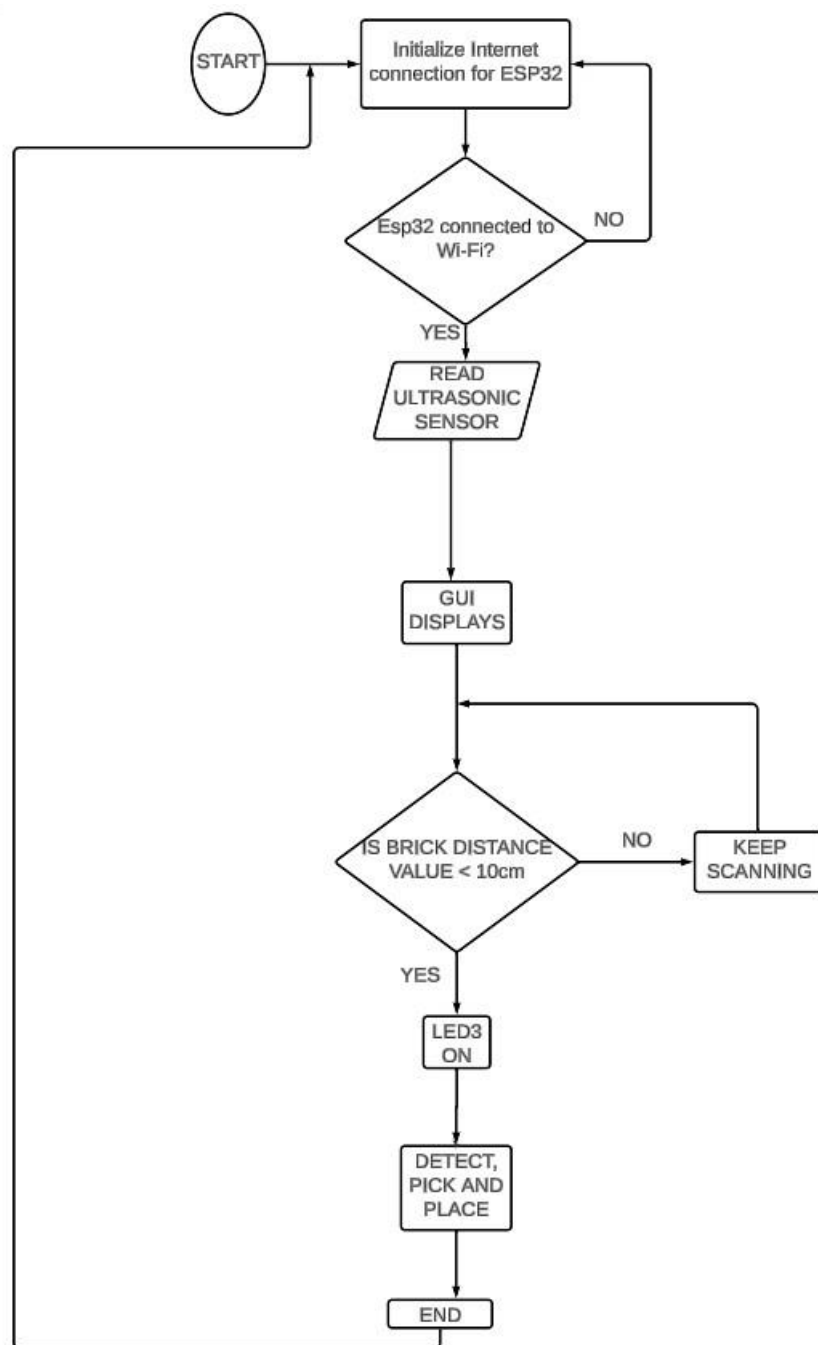


Figure 3. 4: System Flow Chart

The flowchart illustrates how the Construction site robotic arm integrates various components for realtime environmental monitoring and control. The ultrasonic sensor and LDR sensor gather data (distance value, detection status and darkness level), which is then processed by the ESP32 module which is then transmitted wirelessly using Wi-Fi communication to the Blynk app to be displayed on the GUI and for cloud storage. The Blynk GUI allows users to monitor real-time distance value between

the brick and robotic arm, detection status and darkness level, and then compare distance values between the brick and the robot that will enable the robotic arm to pick and place the brick to the truck according to the detected distance value. If the detected distance value between the robotic arm and the brick is less than 10cm, the robot will pick and place the brick to the truck while displaying "BRICK DETECTED" on the GUI and activates the indicating LED to indicate that the brick is detected. If the distance between the brick and the robotic arm is equal to or greater than 10cm there would be no detection of the brick and the robotic arm will continue scanning while displaying, "NO BRICK DETECTED" on the GUI.

CHAPTER 4: CONSTRUCTION

4.1 Electronic Construction

The electronic construction began with the ESP32 microcontroller, which served as the main control unit on the breadboard. All peripherals were connected to the ESP32 via jumper wires.

- Power supply: The ESP32 was powered by a 5V USB connection from a laptop, while the four servo motors were driven by an external regulated 5V DC source for sufficient current. To create a common reference point, the ESP32 ground (GND) was connected to the external supply ground.
- The HC-SR04 ultrasonic sensor was mounted on the side of the robotic arm. Its trigger and echo pins were attached to ESP32 GPIOs 5 and 18, respectively. The sensor was powered by the 5V of the breadboard and GND wires.
- Four servo motors were connected to the ESP32 via GPIO 13 (base rotation), GPIO 12 (shoulder), GPIO 14 (elbow), and GPIO 27 (gripper). Each servo was linked to the external 5V rail and operated using independent PWM signals provided by the ESP32
- Three LEDs were used to show system states: Automatic Mode (GPIO 25), Manual Mode (GPIO 26), and Brick Detected (GPIO 33). To prevent overcurrent from the ESP32 pins, each LED was wired in series with a 220 Ω resistor.

All connections were created on a breadboard, allowing for simple adjustments and testing flexibility. Color-coded jumper wires made it easy to distinguish between power, ground, and signal connections during troubleshooting.

4.2 Mechanical Construction

The mechanical construction of the robotic arm was built with lightweight materials and servo clamps to secure each motor. The arm comprised of four major joints:

- Base Servo: Provides horizontal rotation for the entire arm.
- The shoulder servo controlled the vertical elevation of the arm.
- Elbow Servo: Extend or retract the arm to reach objects.
- The gripper servo opened and closed the claw to hold the brick.

The arm components were secured to a solid base platform to provide optimal balance and motion stability. An ultrasonic sensor was connected near the base to continuously measure the distance between the arm and nearby objects. The mechanical device was created to imitate construction-site lifting and placement procedures on a small scale.



Figure 4. 1: Mechanical Construction of the System

4.3 Final Circuit Prototype Development

The circuit prototype for the Construction Site Robotic Arm was created and constructed on a breadboard. The ESP32 microcontroller, ultrasonic sensor, indicator LEDs, and four servo motors were linked via jumper wires. The ESP32 was powered by the 5V USB supply of the laptop, and the servo motors were driven by an external regulated 5V DC power source to ensure smooth operation. A common ground was shared between the ESP32 and the external power supply to ensure a constant reference voltage and signal integrity across all components.

The circuit enabled bidirectional Wi-Fi connection between the ESP32 and the Blynk Cloud Platform, allowing for remote control as well as real-time monitoring. During operation, the ultrasonic sensor continuously monitored item distance and relayed data to the ESP32, which determined whether to start the autonomous pick-and-place process or remain idle. Manual control was implemented using the Blynk interface, with virtual pins for each servo motor. During testing, the complete system worked as expected, verifying both hardware connections and control logic for real-world building applications.

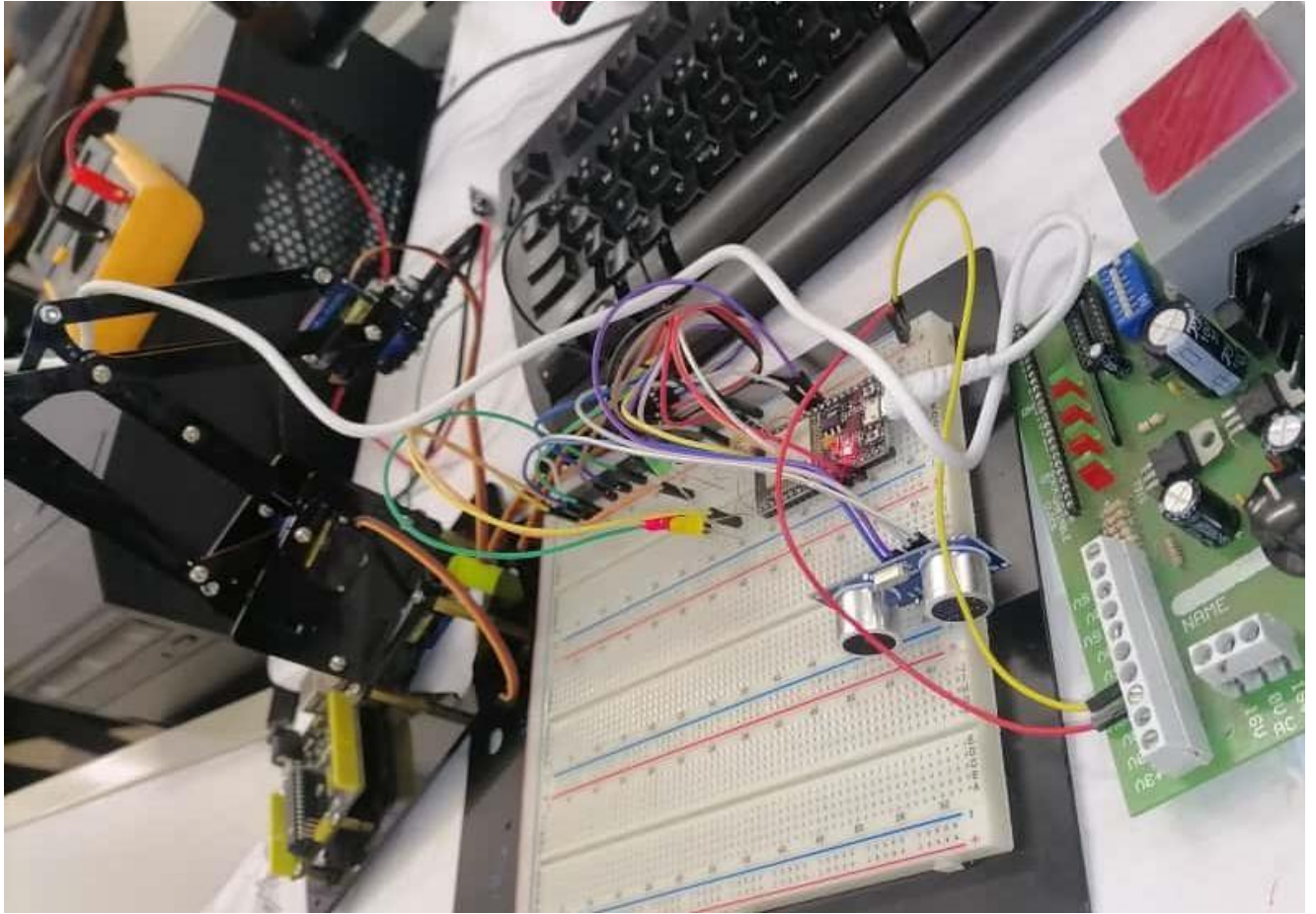


Figure 4. 2: Prototype Of The System

4.4 PROJECT HOUSING

The Construction Site Robotic Arm prototype was built on a small wooden base that was meant to resemble a real-world work site. The arm is securely fixed on an elevated platform to ensure stable movement during operation while preventing vibration transfer to the working area. The ESP32 microcontroller, power supply unit, and other electronic components are located in the upper right side of the base for easy access and cable management. Three indication LEDs are positioned on the left hand side panel to show system status: automated mode, manual mode, and object detection. An ultrasonic sensor is mounted on the right front side of the robotic arm, measuring forward distance for object detection and autonomous operation. The working space includes a tiny construction set with a model dump truck and an imitation brick wall that simulates real-world material handling. The complete arrangement is powered by a 5V regulated power supply, with servo motors stabilized by electrolytic capacitors to reduce noise and vibration. This housing provides an accurate version of an IoT-enabled autonomous robotic arm used in small-scale building applications.

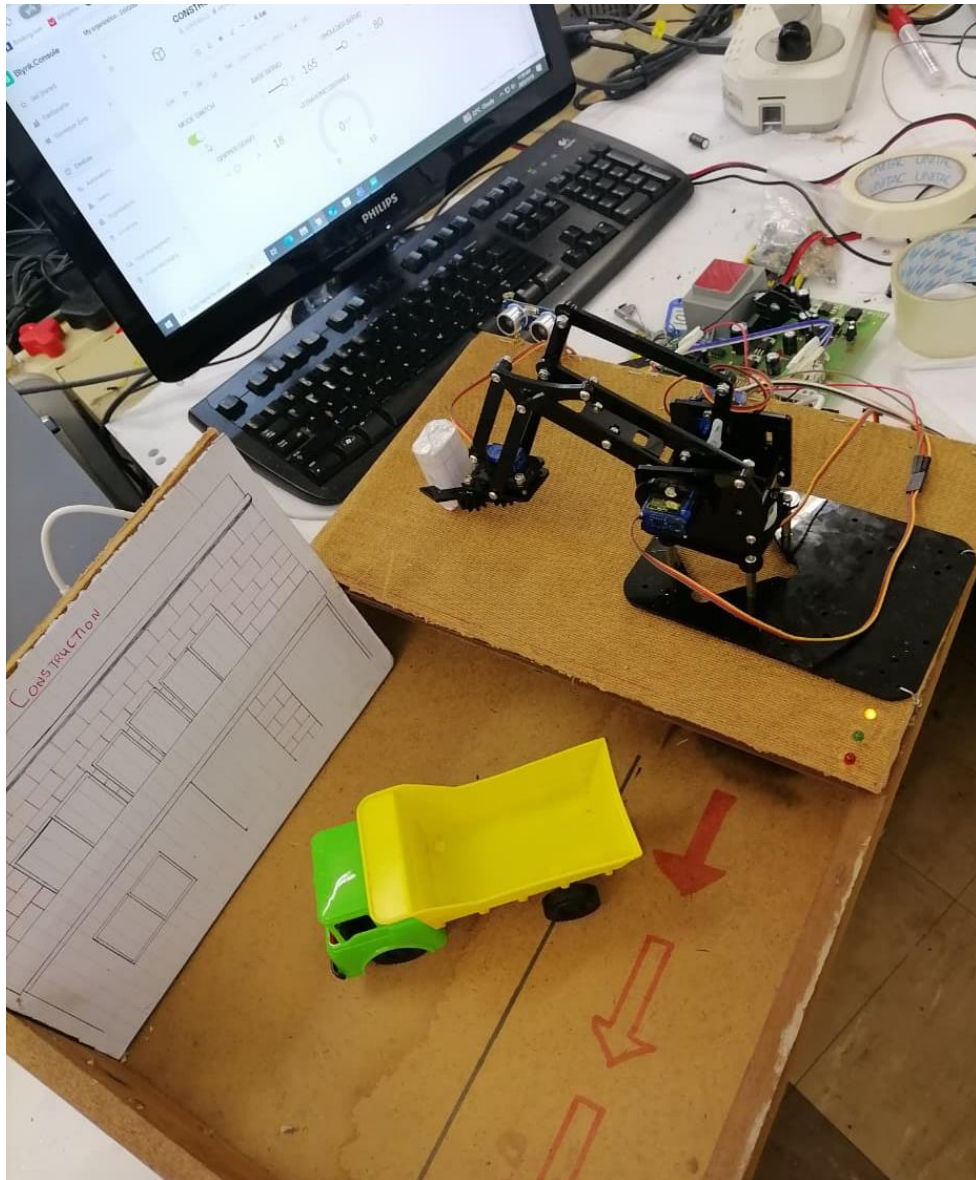


Figure 4. 3:Housing of the Project

CHAPTER 5: TESTING

5.1 Blynk Account Setup and Configuration

The setup and configuration of the Blynk platform served as the initial stage of testing. A new Blynk template was developed for the Construction Site Robotic Arm, with virtual data streams assigned to control the four servo motors and monitor the working modes. The mode switch (Automatic/Manual) was allocated to virtual pin V0, while the manual control buttons for the base, shoulder, elbow, and gripper servos were assigned to V1, V2, V3, and V4, respectively. The ultrasonic sensor values were taken at virtual pin V5. The dashboard featured buttons, sliders, and indication LEDs. The system was tested to confirm that the dashboard responded appropriately to user inputs and got real-time response from the ESP32.

5.2 Communication Between the Blynk Application and the ESP32 Microcontroller Testing

The ESP32 microcontroller was tested to ensure its connectivity, reaction to Blynk commands, and Wi-Fi communication capabilities, before being integrated into the robotic arm system. To execute this test, an LED was connected to one of the GPIO pins of the ESP32 board. The appropriate board drivers were installed to ensure compatibility between the ESP32 and the laptop, allowing the Arduino IDE to detect and upload code to the device. The Blynk and Wi-Fi libraries were then installed and configured to allow connection with the Blynk cloud platform. A virtual button widget was generated in the Blynk app and linked to the same virtual pin as defined in the ESP32 code. The test process involves pressing a button on the Blynk interface to transmit ON/OFF commands to the ESP32, which then controlled the LED state accordingly. This arrangement replicated a basic control signal test to check the ESP32's GPIO output and wireless connectivity functionalities worked properly.

5.3 Wi-Fi and Cloud Communication

Wi-Fi connection between the ESP32 and the Blynk Cloud. The device was set up with the right SSID, password, and Blynk authentication token. Following code upload, the serial monitor reported "Blynk Connected," indicating successful contact with the cloud. Continuous operation testing demonstrated that the ESP32 maintained a stable Wi-Fi network with few connection drops, allowing for consistent data flow between the robotic arm and the user interface.

5.4 Ultrasonic Sensor Testing

The HC-SR04 ultrasonic sensor was evaluated for accuracy in distance measuring. Objects (representing bricks) were placed at various distances, and readings were verified using the serial monitor. The sensor effectively recognized objects in the 2-30 cm range, with the optimal detection point set at 10 cm. The performance of the sensor was consistent under indoor lighting, albeit there were rare false readings with reflecting surfaces. This test proved that the sensor was ready for automatic detection during operation.

5.5 Servo Motor Testing

Each of the four servo motors (base rotation, shoulder lift, elbow reach, and gripper control) was tested separately. The servos were powered by an external 5V regulated source and controlled via ESP32 PWM ports. Movement accuracy, range, and synchronization were tested. The servos operated effectively, and full angular motion was achieved in manual mode using the Blynk interface. Minor jittering was seen when all servos ran concurrently due to the large current draw, which was mitigated by assuring good grounding and a stable power supply.

5.6 Indicator LED Testing

Three LEDs were evaluated as system indicators:

- LED1: Automatic mode indicator.
- LED2: Manual mode indicator.
- LED3: Brick detecting indicator.

Each LED was checked for proper activation based on the programmed system statuses. LEDs responded promptly to mode switch and ultrasonic sensor detections, proving the visual feedback functionality of the system and responsiveness.

5.7 Prototype Manual Control Using Blynk Testing

Manual control was attempted by operating the robotic arm using the Blynk mobile app. Each servo motor accurately responded to the virtual control widgets. The user could rotate the base, lift or lower the arm, stretch the elbow, and open or close the gripper with little delay. The manual mode gave the user entire control over the movement of the arm, proving dependable IoT-based communication and control between the ESP32 and the Blynk app.

5.8 Prototype Automatic Mode Testing

In automatic mode, the ultrasonic sensor recognized an object (brick) at a distance of less than 10 cm and initiated the predetermined pick-and-place process. The base servo spun toward the object, the shoulder and elbow positioned the arm, and the gripper closed to lift the brick. The arm then rotated to release the block at the selected drop position. Throughout testing, the process ran smoothly and consistently, showing that the autonomous operation logic was working properly.

5.9 Dual-Mode Switching (Automatic/Manual)

To test the dual-mode operation of the system, we toggled the mode switch on the Blynk dashboard. In Automatic Mode, the arm ignored manual commands and relied entirely on sensor information. In Manual Mode, the operator regained direct control of the servos. The transitions between modes were flawless, with no command overlap or interruption. This test demonstrated that the system could safely and effectively switch between autonomous and user-controlled operation.

5.10 Final Project Testing with Project Housing

Following successful testing on the breadboard, the entire Construction Site Robotic Arm system was installed in its final enclosure for integrated testing. The housing was intended to resemble an actual construction site, complete with a designated pickup and placing area. The ESP32 microprocessor, power supply unit, and servo motors were securely mounted on the wooden platform to reduce vibration and increase mechanical stability during operation. The ultrasonic sensor was mounted on the front part of the arm to detect bricks within a 10-cm range. The left hand side panel had three indicator LEDs for real-time system feedback: automatic mode, manual mode, and block detection.

During testing, the system was supplied by a regulated 5V supply, and capacitors were installed across the power rails to stabilize current flow to the servo motors, therefore decreasing jitter and undesirable oscillations. In manual mode, control commands were transmitted from the Blynk app, allowing each servo motor to move independently for base rotation, shoulder, elbow, and gripper control. The arm successfully picked up and placed a block in the simulated truck bed. In automated mode, the ultrasonic sensor recognized an object, initiated the automatic pick-and-place process, and released the brick at the drop-off site.

The final prototype testing confirmed full system operation, dependable wireless connectivity, and increased power stability over the breadboard prototype. The dwelling layout demonstrated IoT-enabled automation for small-scale construction applications, resulting in both functional and aesthetic success.

5.10.1 Final Project Manual Mode Testing

Manual mode testing was carried out to ensure direct control of the robotic arm using the Blynk application. Each servo motor (base, shoulder, elbow, and gripper) was tested separately using specific sliders or buttons on the Blynk interface. The ESP32 microprocessor received commands over Wi-Fi, and the servo motors responded in real time. During this test, the system's Manual Mode LED lit up, indicating that it was under user control. The gripper's action was tested for open and close positions (ranging from 5° to 90°), and the arm was manually positioned to pick up and deposit a small simulated brick in the desired location. The ultrasonic sensor was also active, although it was not used to initiate movement in this mode. This test was designed to confirm the responsiveness, precision, and smooth operation of each joint under user control.

5.10.2 Final Project Automatic Mode Testing

Automatic mode testing assessed the ability of the arm to conduct the pick-and-place sequence autonomously using sensor input. The Automatic Mode LED illuminated, indicating that the system was running without manual intervention. The ultrasonic sensor was calibrated to detect an object within 10 cm. When an object was spotted, the Detection LED lit up, and the ESP32 activated the automatic pick-and-place function. The robotic arm followed this sequence:

1. Rotate base toward detected object.
2. Lower shoulder and elbow to approach the object.
3. Close the gripper to hold the object securely.
4. Lift the object slightly and rotate toward the placement area.
5. Lower the arm and open the gripper to release the object.
6. Return to the middle (ready) position.

Testing demonstrated that the arm completed each step in the correct order, and that the system was capable of detecting, picking, and placing things repeatedly under steady power supply conditions.

CHAPTER 6: RESULTS

6.1 Blynk Account Setup and Configuration

The Blynk account was successfully set up for the Construction Site Robotic Arm. The virtual data streams were assigned as follows: V0 for mode selection (Automatic/Manual), V1-V4 for servo motor control, and V5 for ultrasonic distance measurements. Each widget, including buttons and sliders, was correctly attached to the associated virtual pin. The dashboard responded quickly, providing live distance readings and servo movement updates. The setup confirmed that cloud communication and interface mapping were fully operational.

6 Datastreams

☐	ID	Name	Pin	Color	Data Type	Units	is	Actions
☐	1	MODE SWITCH	V0		Integer		false	
☐	2	BASE SERVO	V1		Integer	°	false	
☐	3	SHOULDER SERVO	V2		Integer	°	false	
☐	4	ELBOW SERVO	V3		Integer	°	false	
☐	5	GRIPPER SERVO	V4		Integer	°	false	

Figure 6. 1: The Setup And Configuration Of The Blynk Platform

6.2 Communication Between the Blynk Application and the ESP32 Microcontroller Results

The ESP32 successfully connected to a Wi-Fi network and communicated with the Blynk cloud server. When the virtual button on the Blynk app was pressed, the LED of the ESP32 turned on; when the button was released or toggled off, the LED shut off. The LED reaction was immediate, indicating realtime communication between the Blynk application and the ESP32 microcontroller. This indicated that the ESP32 was properly functional, reacting reliably to user requests, and capable of establishing a stable Wi-Fi connection. The test confirmed that the ESP32 was ready for integration with other components such as the ultrasonic sensor and servo motors.

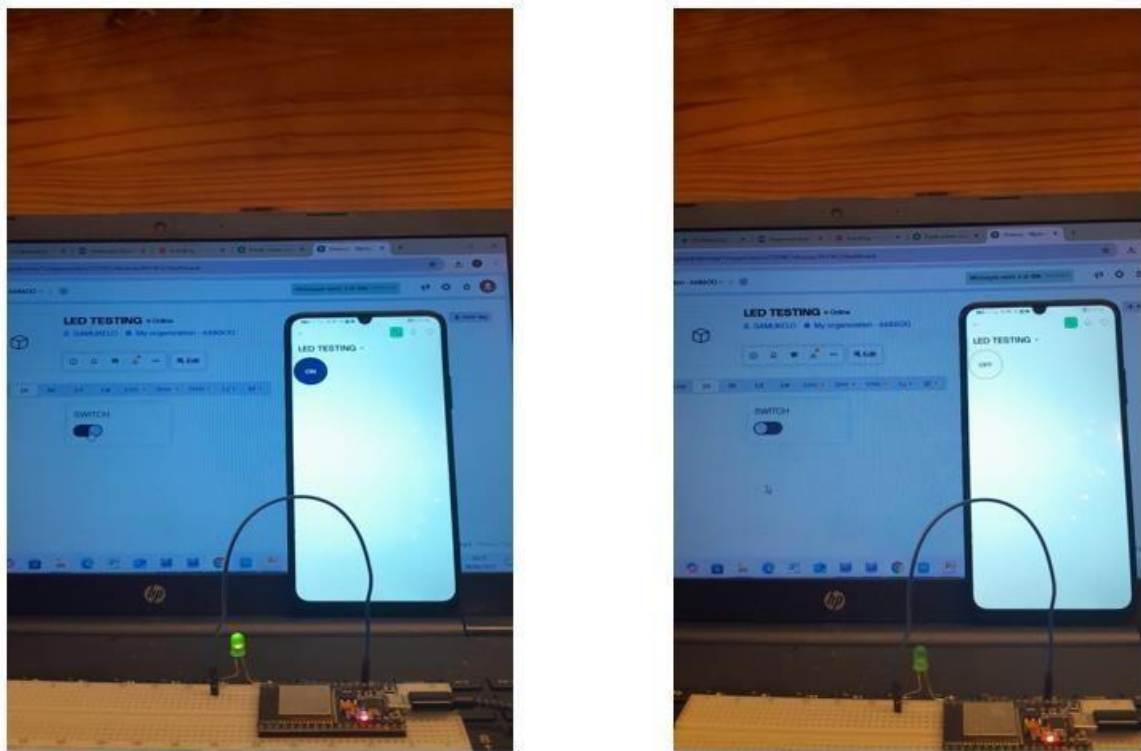


Figure 6. 2: Communication Between the Blynk Application and the ESP32 Microcontroller

6.3 Wi-Fi Communication

After powering on the ESP32, the serial monitor displayed "Blynk Connected," indicating a successful Wi-Fi and cloud connection. The ESP32 maintained an efficient connection throughout testing, with no significant dropouts or data transmission delays. This guaranteed that the robotic arm and the Blynk IoT platform communicated smoothly and in real time, allowing for dependable remote operation.

6.4 Ultrasonic Sensor Readings

The HC-SR04 ultrasonic sensor accurately recognized objects in the 2-65cm range. The optimal reaction distance was measured at 10 cm, at which point the "Brick Detected" condition was activated. The Blynk interface displayed the real-time readings, which updated quickly as the object moved closer or away. This confirmed that the sensor and ESP32 integration was successful, resulting in precise and responsive distance measurements.



Figure 6. 3:Ultrasonic Sensor Readings

6.5 Servo Motor Control Results

Four servo motors were tested individually: base rotation, shoulder, elbow, and gripper. During manual operation, each servo responded promptly to the virtual control supplied to it on Blynk. Even in automatic mode, the motors displayed consistent and synchronized action. A slight vibration occurred when all motors moved at the same time, which was reduced by utilizing a regulated external 5V supply. Overall, the motion control system performed as planned in terms of reliability and accuracy.

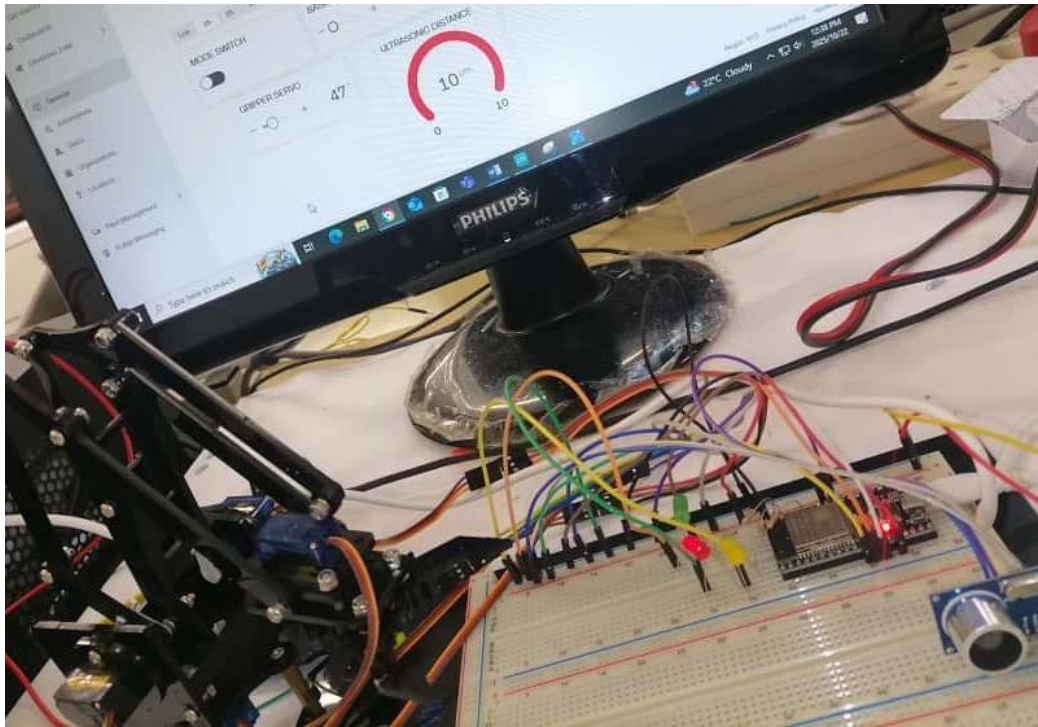


Figure 6. 4: Servo Motor Control Results

6.6 Indicator LED Operation

Three indicator LEDs worked just as designed:

- Yellow LED (Automatic Mode) turns on when the system enters automatic mode.
- Red LED (Manual Mode) turns on when manual control is activated.
- Green LED (Brick Detection) turns on when an ultrasonic sensor detects an object within 10 cm.

The LEDs gave clear visual status input, which improved user awareness of system operation.



Figure 6. 5: Indicator LED Operation Results

6.7 Prototype Manual Mode Operation Results

In manual mode, the robotic arm accurately responded to user commands sent through the Blynk interface. Each servo could be operated individually to rotate the base, raise or lower the arm, extend or retract the elbow, and open or close the gripper. The red LED came on to indicate the manual mode and commands were executed in real time with low latency, demonstrating the responsiveness of IoT connectivity. This demonstrated complete user control over the robotic arm, enabling precise material handling.

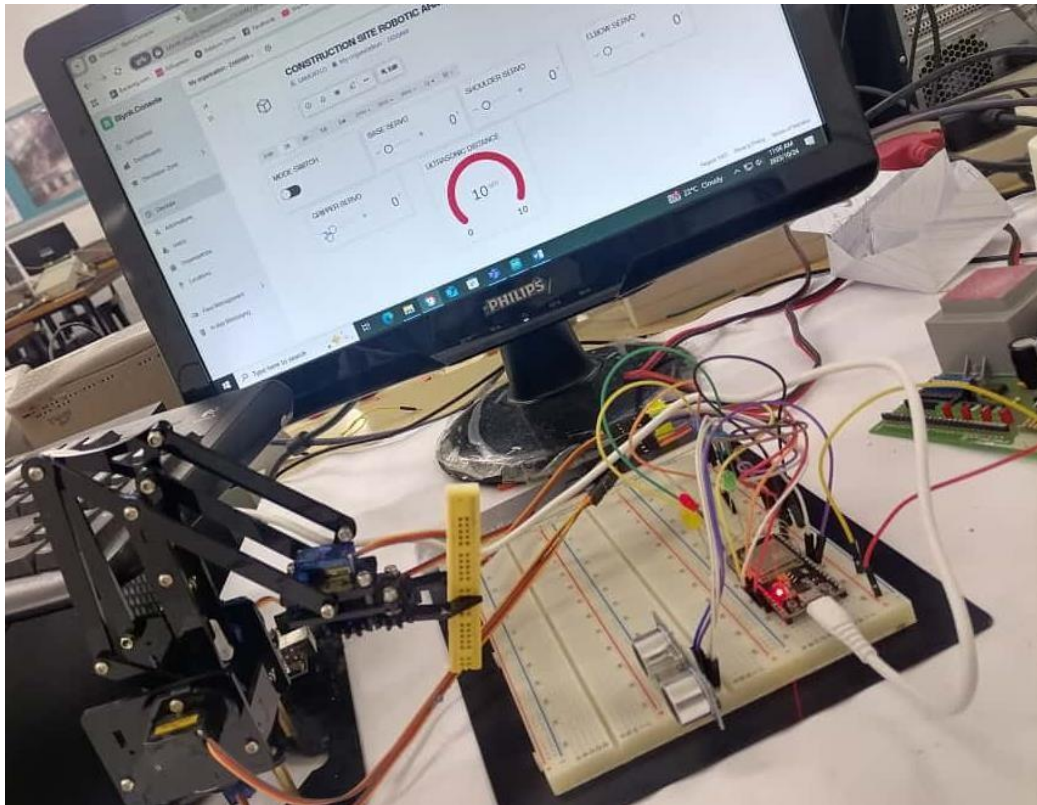


Figure 6. 6: System Operating In The Manual Mode

6.8 Prototype Automatic Mode Operation Results

In automatic mode, the ultrasonic sensor served as a trigger for robotic activity. When the distance reading was less than 10 cm, the ESP32 carried out the programmed pick-and-place sequence:

1. The base rotated towards the object.
2. Aligned the gripper using the shoulder and elbow.
3. Close the gripper to pick up the brick.
4. The arm lifted, spun, and released the brick at the designated drop location.

All actions were completed sequentially and without error, demonstrating that the programmed automation logic was working as intended. The yellow LED came on indicating the automatic mode and

since the distance between the ultrasonic sensor and the object was less than 10cm as displayed the in the blynk GUI the reading of 9cm which automatically activated the green LED indicating that the object has been detected.

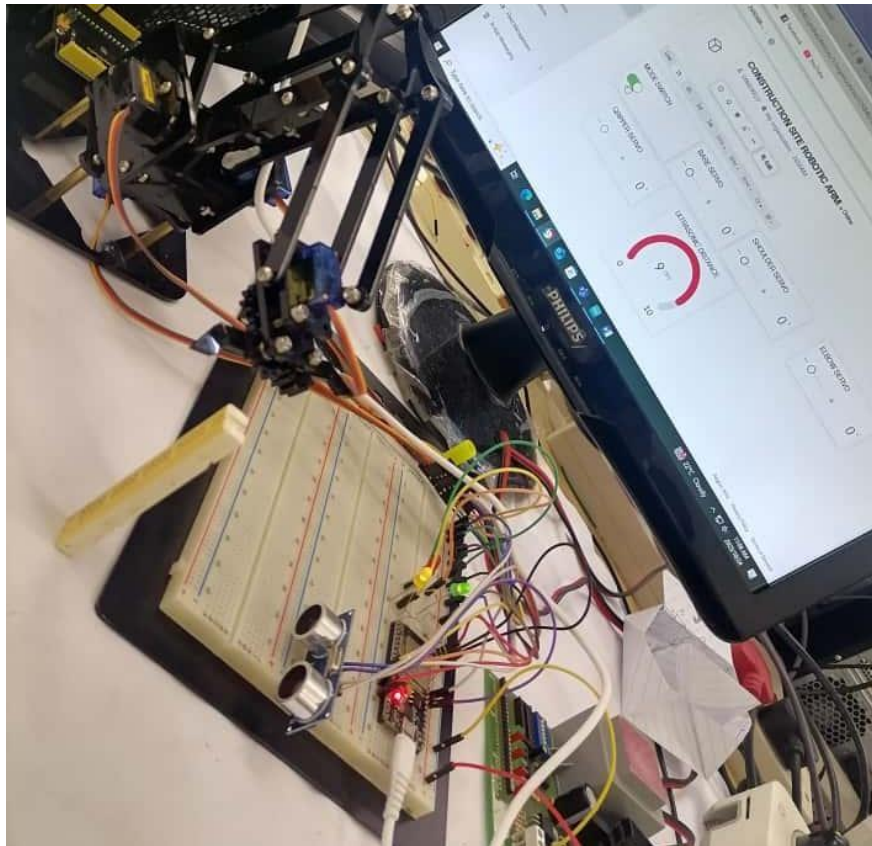


Figure 6. 7: System Operating In The Automatic Mode

6.9 Dual-Mode Switching

The switch between Automatic and Manual modes using Blynk virtual pin V0 was successful. When Automatic mode was enabled, manual controls were disabled to prevent disturbance. When the user switched to Manual mode, they regained full control. Mode transitions were smooth and fast, with LED indicators updating in real time. This verified the correct dual-mode control logic implementation.



Figure 6. 8: Dual-Mode Switching Results

6.10 Final Project Results with Project Housing

The final prototype testing demonstrated that the Construction Site Robotic Arm effectively performed both manual and automatic functions as expected. In manual mode, the Blynk interface provided precise control over each servo motor, demonstrating effective bidirectional communication between the ESP32 microcontroller and the Blynk cloud. The arm moved smoothly and responsively, with minimum latency detected during control actions. In automatic mode, the ultrasonic sensor accurately recognized objects within a 10 cm range and initiated the pre-programmed pick-and-place procedure. The system's indicator LEDs gave clear visual input on the operating mode and detection status, facilitating human involvement and monitoring.

The use of capacitors across the servo power rails successfully reduced motor vibration and instability, resulting in smoother operation than the breadboard prototype. The power supply performed consistently during testing, and the ESP32 maintained a reliable Wi-Fi connection with no disconnections or delays. The physical housing increased the mechanical strength of the system while reducing noise and movement interference during arm operation. Overall, the results verified that the system met its design goals, displaying dependable IoT-based control, precise sensing, and realistic automation appropriate for small-scale construction settings.

6.10.1 Final Project Manual Mode Results

During manual operation, all four servos responded precisely to control inputs from the Blynk app. The robotic arm moved precisely and smoothly in each axis, and the Manual Mode LED clearly displayed the operational condition. The gripper successfully picked up and placed the simulated block several times without slipping. The results verified that the manual mode provided consistent user control, precise servo response, and low power fluctuation.

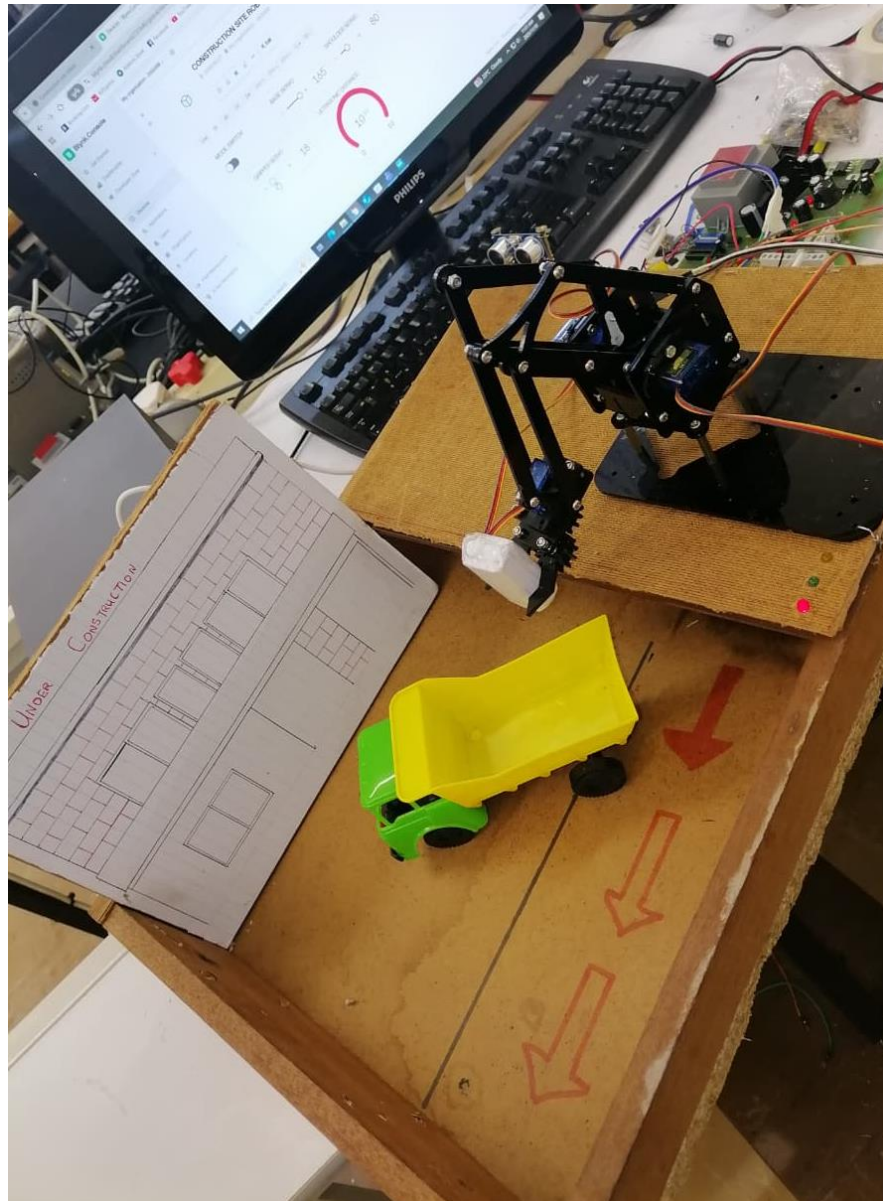


Figure 6. 9:Final Project Manual Mode Results

6.10.2 Final Project Automatic Mode Results

In automatic mode, the ultrasonic sensor accurately recognized objects within the designated 10 cm range, activating the Detection LED and starting the automatic sequence. The robotic arm successfully turned, lowered, grasped, lifted, and placed the brick in its intended location. The Automatic Mode LED stayed illuminated throughout the cycle, indicating that the device was operating autonomously. The pick-and-place motion was smooth, consistent, and stable. Capacitors across the servo power rails successfully minimized jitter, resulting in smooth operation throughout the sequence. Multiple experiments yielded consistent results, demonstrating the efficiency of programmed automation.

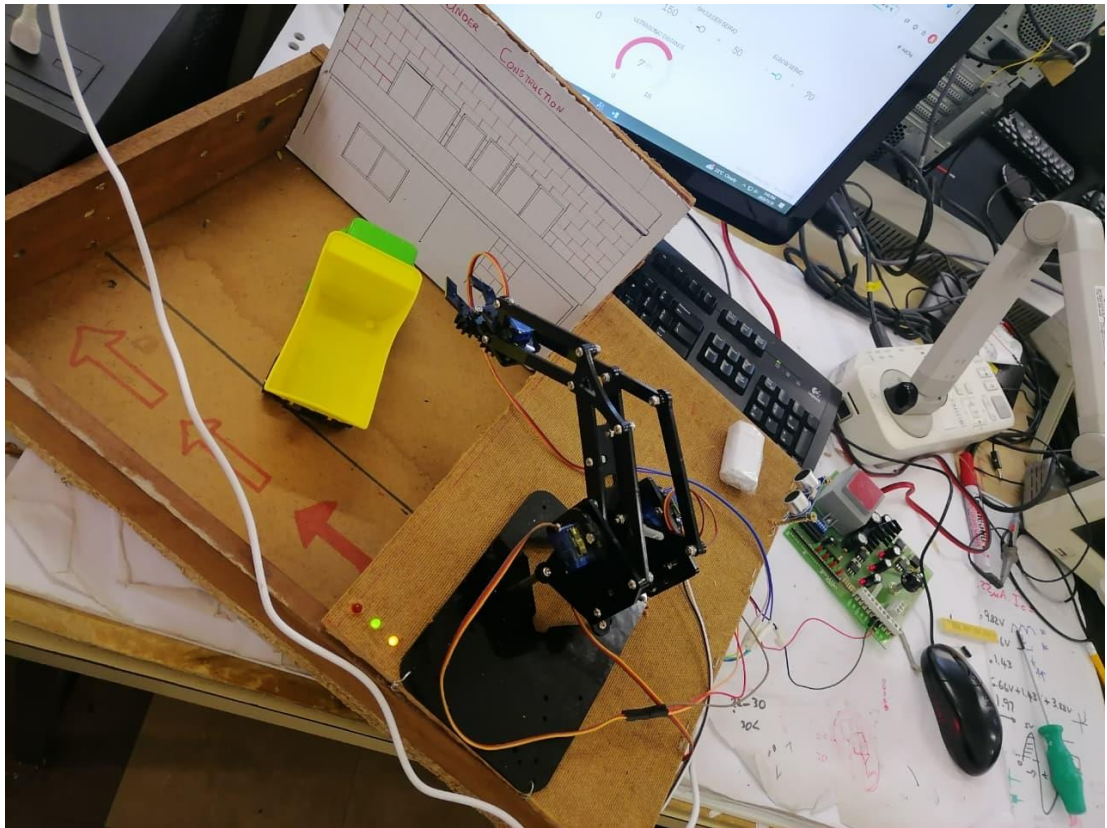


Figure 6. 10: Final Project Automatic Mode Results

6.11 Performance Evaluation

Table 6.1 shows performance evaluation for the results analysed. It summarizes the performance of the Construction Site Robotic Arm by comparing expected and actual results during testing.

Table 6. 1: Performance Evaluation Table

Test Parameter	Expected Result	Observed (Actual) Result	Remarks
Wi-Fi Connectivity (ESP32)	Stable connection to Blynk app via Wi-Fi	Connection remained stable throughout testing	Successful – no disconnections observed
Manual Mode Operation	User to control all four servo motors individually via Blynk sliders	All servos responded smoothly to manual commands	Functionality achieved – full manual control verified
Automatic Mode Operation	Automatic detection and pick-and-place sequence triggered when object < 10 cm	Detection successful; automatic pick-and-place executed as programmed	Fully functional – sequence runs reliably
Object Detection Accuracy (Ultrasonic Sensor)	Detect objects within 10 cm threshold	Detected object between 4–10 cm accurately	Sensor accurate within set range
Mode Indicator LEDs	Auto, Manual, and Detection LEDs show corresponding operation state	LEDs correctly displayed each system mode	Clear and reliable indication achieved
Servo Power Stability	Stable 5 V operation with minimal noise	Capacitors reduced jitter and ensured stable motion	Improvement observed after capacitor integration
Gripper Mechanism	Proper open and close response	Gripper opens/closes correctly with angle range 5°–90°	Accurate control achieved
System Response Time	Immediate response to Blynk commands	Delay less than 1 second	Real-time response achieved
Power Supply Performance	5 V regulated supply sufficient for all servos	Stable power with no voltage drops	Consistent supply maintained
Overall System Functionality	Seamless operation in both modes	Fully functional with stable motion and sensing	System meets design objectives

CHAPTER 7: TIME MANAGEMENT

7.1 GANTT CHART

Figure 7.1 depicts the Gantt chart used in this project to convey the project plan, reduce resource overload, track progress, complete projects ahead of schedule, and boost productivity.

7.2. GOLDEN TRIANGLE PLAN

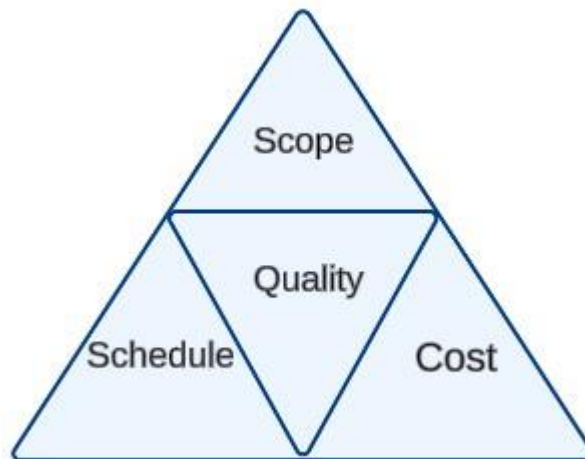
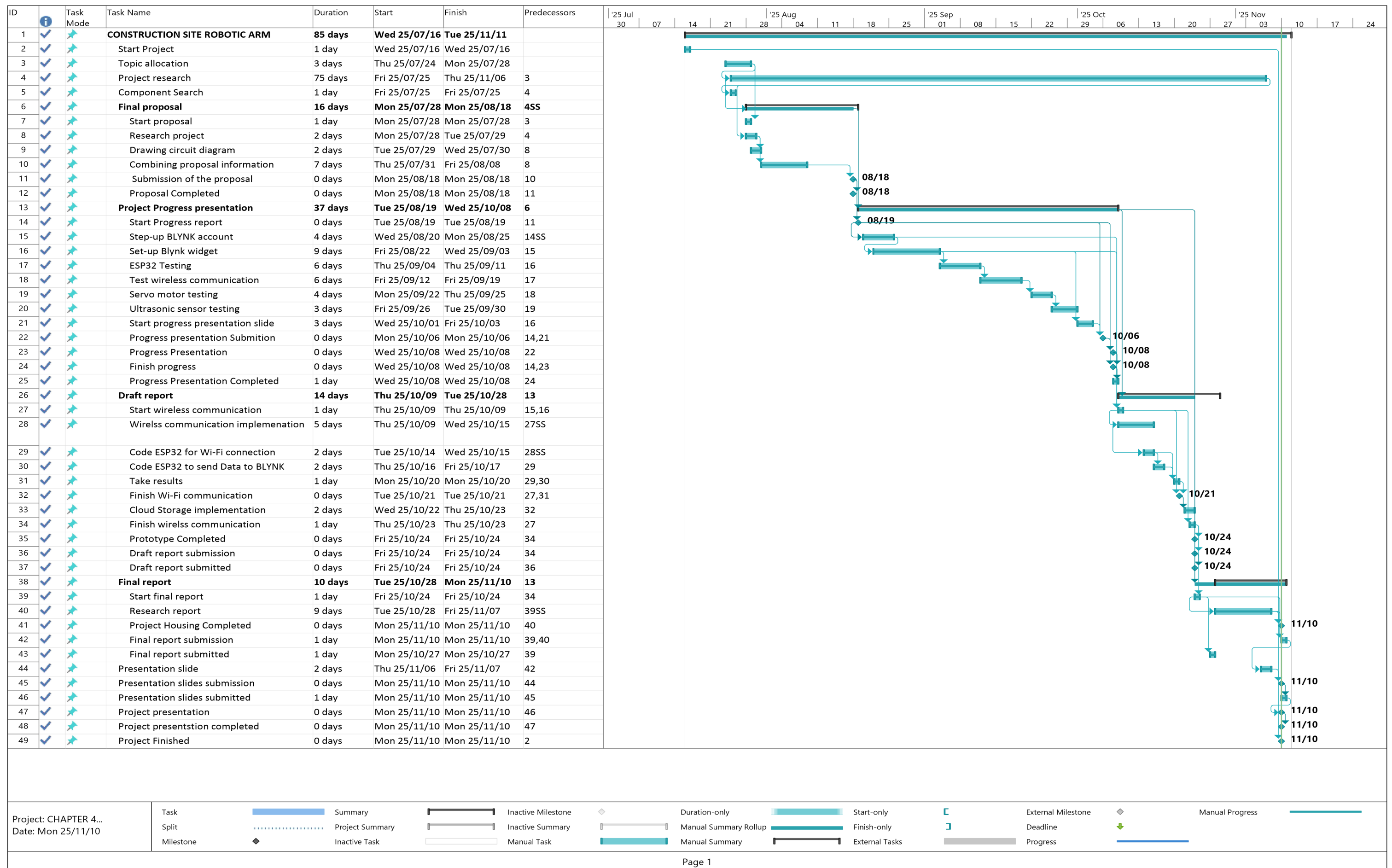


Figure 7. 1: System Golden Triangle

Figure 7.1 shows the golden triangle, which describes the success of the project. The main goal is to get the project done well. The excellence of the project will be achieved by balancing scope, cost, and schedule. Any one of the three being out of balance could lead to a bad or failed project.

- **Scope Definition:** A clear and straightforward project scope will be developed, with the primary goal of developing a construction site robotic arm. The scope definition facilitated effective task planning, organization, and execution.
- **Scheduling Performance:** A detailed project schedule will be produced to outline task sequences and dates. The Gantt Chart (Figure 4.1) shows that the activities of the project were successfully completed on time or ahead of schedule. This provided plenty of time for error correction and testing.
- **Cost Control:** A budget of R4000.00 was set up for the project. Effective resource allocation, procurement, and cost control resulted in a project spend of only R2300.00. The project will deliver high-quality results while reducing costs.



CHAPTER 8: POSSIBLE PROBLEMS AND SOLUTIONS

The dependability and performance of the Construction Site Robotic Arm can be influenced by a variety of technical, environmental, and operational factors. By anticipating these difficulties and taking preventative measures, the system can become more dependable and efficient while maintaining the safety of operators and equipment.

8.1 ENVIRONMENTAL FACTORS

8.1.1 Harsh Construction Site Conditions

Problem: Prolonged exposure to dust, moisture, and vibration on a building site can harm sensors and motors, reducing their functionality.

Solution: Protect critical components like the ESP32, motor drivers, and sensors with dust and moisture-resistant housings. Apply conformal coating on PCBs, and make sure all connections are protected. To preserve accuracy, do periodic sensor inspections and calibrations.

8.1.2 Power Supply Stability

Problem: On construction sites, unreliable or interrupted power sources might disrupt the operation of the robotic arm, perhaps resulting in unplanned stoppage during lifting activities.

Solution: Use a separate regulated DC power source or rechargeable battery pack that can produce a consistent voltage and enough current to power numerous servo motors at once. Use a voltage regulator to keep fluctuations from damaging electronics.

8.2 HEALTH AND SAFETY RISKS

8.2.1 Availability and Fatigue of the operator

Problem: When a key operator or technician is missing due to illness or weariness, progress might be slowed, particularly during the testing or deployment stages.

Solution: Cross-train team members to use and troubleshoot the robotic arm. Maintain open communication throughout the team to transfer tasks as needed.

8.2.2 Potential hazards

Problem: Excessive heat from servo motors or electrical components can cause burns, equipment damage, and even fires.

Solution: Design the arm with adequate ventilation and airflow. To prevent overheating, add thermal monitoring to essential components and programmatically establish automated shutdown triggers.

8.3 IoT & Communication Issues

8.3.1 Wireless Control Interference

Problem: Wi-Fi interference on construction sites can create delays in control signals from the Blynk app to the ESP32, affecting arm reaction times.

Solution: Use dual-core capacity of the ESP32 to segregate communication and motor control activities. Implement signal buffering and retry methods to ensure that commands are carried out without error.

8.3.2 Unauthorized Control Risks

Problem: Unauthorized use of the robotic arm due to malicious access to the mobile app poses a safety concern.

Solution: Enable secure login authentication in the Blynk app and encrypt data sent between the app and the ESP32 to avoid interception or unwanted access.

8.4 Component Efficiency and Sensor Accuracy

8.4.1 Mechanical Wear and Tear

Problem: Continuous use can wear down servo gears, connections, and gripper mechanisms, resulting in decreased performance.

Solution: Use high-quality metal-gear servos like the MG996R, lubricate them regularly, and replace worn parts as soon as possible. To prevent loosening due to vibration, use robust soldering and connectors.

8.4.2 Sensor inaccuracy

Problem: The ultrasonic sensor or load cell may produce misleading readings as a result of dust accumulation, misalignment, or continuous use.

Solution: To keep sensors accurate, clean and calibrate them on a regular basis. Update the ESP32 firmware to adjust readings depending on calibration values and filter out noise from measurements.

CONCLUSION

The Construction Site Robotic Arm project successfully met the primary goal of creating and implementing an IoT-based robotic system capable of detecting, lifting, and positioning construction items such as bricks. The solution provided reliable connection between the hardware and the user interface by integrating an ESP32 microcontroller, the Blynk IoT platform, an ultrasonic sensor, and servo motors. Both manual and automatic operating modes were successfully tested: in manual mode, servo movements responded properly to user directions via the Blynk app, while in automatic mode, the ultrasonic sensor recognized objects within a 10 cm range and started the pick-and-place sequence.

Key design decisions included incorporating indication LEDs for operating feedback, using an external 5V power source to meet current demands, and adding capacitors to reduce servo jittering. These enhancements increased stability, user awareness, and power reliability. Testing demonstrated consistent Wi-Fi connectivity, accurate distance measurement, and exact control of each servo motor, thereby certifying the functionality of the system.

During installation, various issues were encountered, such as minor ultrasonic false triggers and initial servo instability. These concerns were handled via sensor calibration, correct grounding, and signal filtering. The final prototype worked consistently, demonstrating that low-cost components may successfully mimic industrial automation concepts on a tiny scale.

Overall, the research demonstrated the feasibility of combining IoT technology with robotic systems for construction applications. It also underlined the significance of good electronic design methods, such as power management and component isolation, in achieving reliable operation. Future enhancements could include a mobile base for arm mobility, load sensors for weight sensing, and a camera for better item recognition. The Construction Site Robotic Arm is an effective and scalable example of IoT-based automation with real-world applications in material handling and construction technologies.

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ANNEXURE A

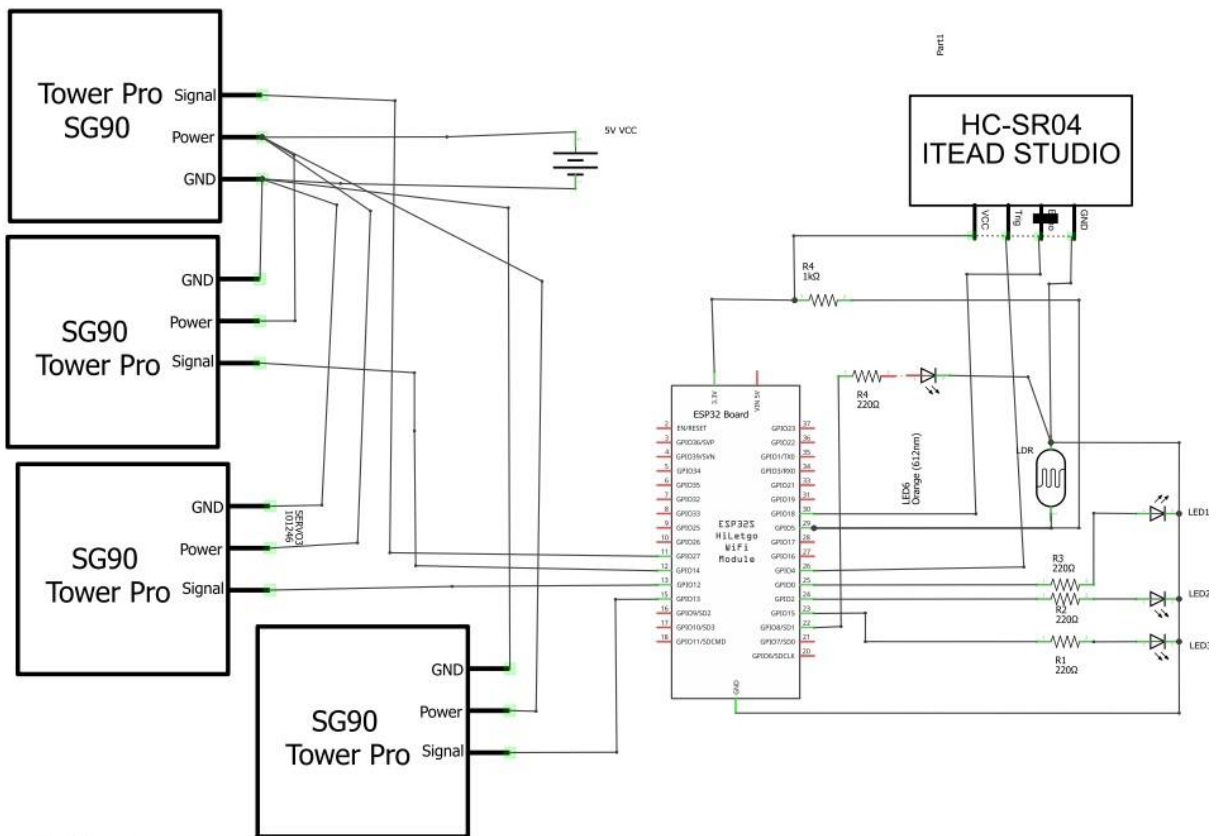


Figure A3. 1: System Schematic Diagram

ANNEXURE B

```
#define BLYNK_TEMPLATE_ID "TMPL2OymXov2g"
#define BLYNK_TEMPLATE_NAME "CONSTRUCTION SITE ROBOTIC ARM"
#define BLYNK_AUTH_TOKEN "oe9DJfDfsNbZ209w1lw1Vk0oEb_CINwu"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h>
#include <ESP32Servo.h>

// ----- Blynk Auth & Wi-Fi -----
char auth[] = "oe9DJfDfsNbZ209w1lw1Vk0oEb_CINwu"; // paste from Blynk
char ssid[] = "SAMUKELO"; // your Wi-Fi SSID
char pass[] = "SAMUKELO117"; // your Wi-Fi password

// ----- Servo Setup -----
Servo servoBase;
Servo servoShoulder;
Servo servoElbow;
Servo servoGripper;

#define BASE_PIN 13
#define SHOULDER_PIN 12
#define ELBOW_PIN 14
#define GRIPPER_PIN 27

// ----- Ultrasonic Setup -----
#define TRIG_PIN 5
#define ECHO_PIN 18

long duration;
int distanceCm;

// ----- LEDs Setup -----
#define LED_AUTO 21
#define LED_MANUAL 22
#define LED_BRICK 23

// ----- Mode Flag -----
bool isAuto = false;

// ----- Blynk Virtual Pin Handlers -----

// Mode switch (V0)
BLYNK_WRITE(V0) {
  int mode = param.asInt(); // 1 = Auto, 0 = Manual
  isAuto = (mode == 1);

  if (isAuto) {
    digitalWrite(LED_AUTO, HIGH);
    digitalWrite(LED_MANUAL, LOW);
  } else {
    digitalWrite(LED_AUTO, LOW);
    digitalWrite(LED_MANUAL, HIGH);
  }
}
```

```

// Manual control for Base (V1)
BLYNK_WRITE(V1) {
  if (!isAuto) {
    int angle = param.asInt();
    servoBase.write(angle);
  }
}

// Manual control for Shoulder (V2)
BLYNK_WRITE(V2) {
  if (!isAuto) {
    int angle = param.asInt();
    servoShoulder.write(angle);
  }
}

// Manual control for Elbow (V3)
BLYNK_WRITE(V3) {
  if (!isAuto) {
    int angle = param.asInt();
    servoElbow.write(angle);
  }
}

// Manual control for Gripper (V4)
BLYNK_WRITE(V4) {
  if (!isAuto) {
    int angle = param.asInt();
    servoGripper.write(angle);
  }
}

// ----- Functions -----

// Measure distance with HC-SR04
int readDistanceCM() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH, 30000); // timeout 30ms
  int distance = duration * 0.034 / 2; // cm
  return distance;
}

// Auto Mode: Pick & Place sequence
void autoPickAndPlace() {
  // Example sequence (adjust angles as needed)
  servoBase.write(80); delay(500);
  servoShoulder.write(60); delay(500);
  servoElbow.write(60); delay(500);
  servoGripper.write(15); delay(500); // close gripper
  delay(1000);
  servoShoulder.write(50); delay(500);
  servoBase.write(150); delay(500);
}

```

```

    servoElbow.write(70); delay(500);
    servoGripper.write(90); delay(500); // open gripper
}

// ----- Setup -----
void setup() {
    Serial.begin(115200);

    // Attach servos
    servoBase.attach(BASE_PIN);
    servoShoulder.attach(SHOULDER_PIN);
    servoElbow.attach(ELBOW_PIN);
    servoGripper.attach(GRIPPER_PIN);

    // Setup ultrasonic pins
    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);

    // Setup LEDs
    pinMode(LED_AUTO, OUTPUT);
    pinMode(LED_MANUAL, OUTPUT);
    pinMode(LED_BRICK, OUTPUT);

    // Default LEDs: Manual ON at start
    digitalWrite(LED_AUTO, LOW);
    digitalWrite(LED_MANUAL, HIGH);
    digitalWrite(LED_BRICK, LOW);

    // Connect to Blynk
    Blynk.begin(auth, ssid, pass);
}

// ----- Main Loop -----
void loop() {
    Blynk.run();

    // Read ultrasonic distance
    distanceCm = readDistanceCM();
    Blynk.virtualWrite(V5, distanceCm);

    // Brick detection LED
    if (distanceCm > 0 && distanceCm <= 10) {
        digitalWrite(LED_BRICK, HIGH);
        if (isAuto) {
            autoPickAndPlace();
        }
    } else {
        digitalWrite(LED_BRICK, LOW);
    }
}

```